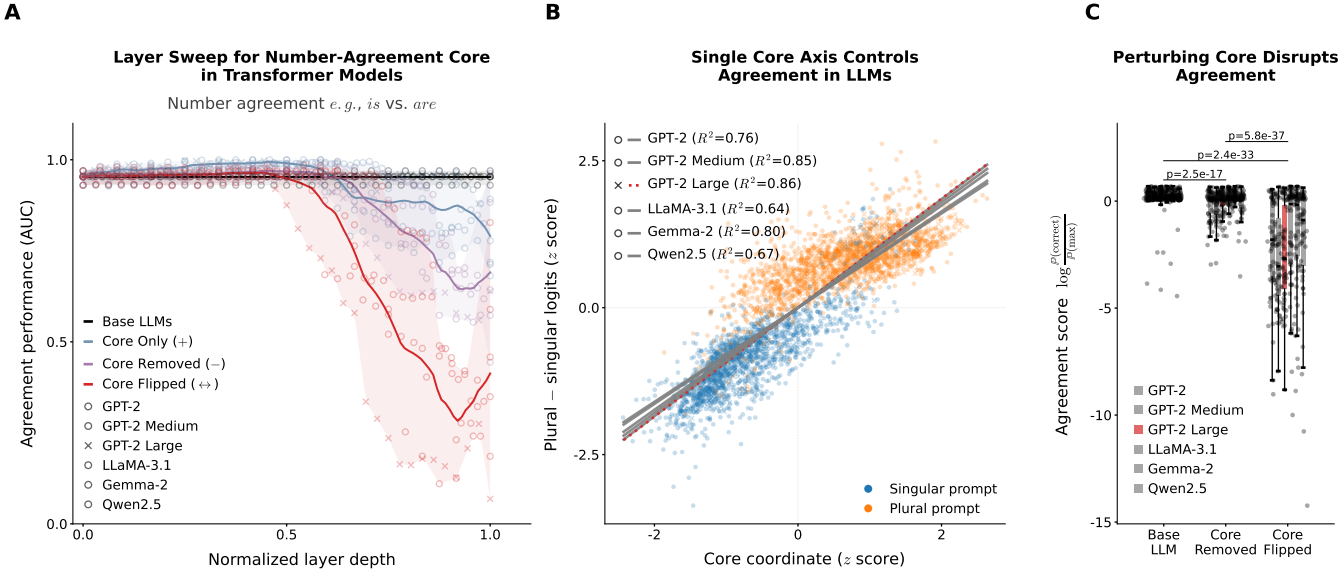

A
Layer Sweep for Number-Agreement Core in Transformer Models
Number agreement e.g., is vs. are
Agreement performance (AUC)
Normalized layer depth
Base LLMs
Core Only (+)
Core Removed (−)
Core Flipped (↔)
GPT-2
GPT-2 Medium
GPT-2 Large
LLaMA-3.1
Gemma-2
Qwen2.5
B
Single Core Axis Controls Agreement in LLMs
Plural − singular logits (z score)
Core coordinate (z score)
GPT-2 (R^2=0.76)
GPT-2 Medium (R^2=0.85)
GPT-2 Large (R^2=0.86)
LLaMA-3.1 (R^2=0.64)
Gemma-2 (R^2=0.80)
Qwen2.5 (R^2=0.67)
Singular prompt
Plural prompt
C
Perturbing Core Disrupts Agreement
Agreement score $\log \frac{P(\text{correct})}{P(\text{max})}$
p=5.8e-37
p=2.4e-33
p=2.5e-17
Base LLM
Core Removed
Core Flipped
GPT-2
GPT-2 Medium
GPT-2 Large
LLaMA-3.1
Gemma-2
Qwen2.5